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Prediction of Storage and Loss Modules of Preformed Particle Gels Using Machine Learning

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Abstract

Gel treatment is a useful technique to reduce water loss and block highly permeable zones along the reservoir. Thus, the rheological properties are crucial for the material to withstand different reservoir and fluid properties. By understanding the viscoelastic properties of preformed particle gels (PPGs) (storage and loss modules), the formulations can be tailored to specific reservoir conditions, optimize the injection strategies, and ensure long-term efficiency in water conformance control and enhanced oil recovery operations. A machine learning approach was developed to predict the storage modulus (G') and loss modulus (G") of PPGs based on key parameters such as PPG type, frequency, temperature, salinity, particle size, and shear strain. Experimental data were collected from published studies on various PPG formulations. The Decision Tree (DT), Support Vector Regression (SVR), Gradient Boosting Regressor (GBR), and XGBoost models were employed. K-fold cross-validation and grid search optimization were applied using the mean squared error and coefficient of determination as performance metrics. The trained models were validated against independent datasets. The comprehensive error analysis, including R-squared values, mean squared errors, and scatter plots, enabled an objective comparison of the predictive models. XGBoost demonstrated superior performance among the individual models, achieving the highest R-squared (99.33% and 98.53%) and lowest mean squared error (0.05 and 0.09). Temperature and salinity had the most substantial impacts, with higher temperatures generally decreasing G' while increasing G". The hyperparameter tuning further refined the model performance, and validation against experimental datasets confirmed the robustness of the predictions. The trend analysis reinforced the reliability of the XGBoost model in capturing the key factors affecting the PPG rheology under reservoir conditions. This predictive capability allows for more precise formulation adjustments, optimizing gel performance for specific field applications. The findings provide a novel tool for designing PPGs tailored to diverse reservoir conditions to predict viscoelastic behavior of PPGs, thereby improving water conformance control strategies.

Introduction

Gel treatment is a traditional, effective, and widely used method of controlling water mobility (Sydansk et al. 2005). In situ gel injection usually exhibits higher resistance factors than polymer flooding. However, field operations revealed some drawbacks, such as damage to low-k zones, poor gel stability, reduced gel strength, and polymer degradation. These challenges make in-depth conformance control, i.e., preformed particle gels, more favorable because of PPG's greater reliability under different conditions (Goudarzi et al. 2015; Yu, Sang, and Dong 2018). PPGs are water-swollen, crosslinked polymer particles. Unlike in situ gelants, PPGs are synthesized and crosslinked at the surface and then injected to block high-k zones (Almakimi et al. 2024; Pi et al. 2023). Over the last few years, experimental studies have focused on improving PPG formulations to withstand harsh reservoir conditions (high temperature and salinity) and to tailor their mechanical properties (storage modulus G' and loss modulus G"), swelling behavior, and particle size to improve performance.

The viscoelastic moduli, such as the storage modulus G' (elastic component) and loss modulus G" (viscous component), are crucial parameters for characterizing the PPG strength and injectivity. Recent experimental rheology studies have revealed that most PPGs behave as elastic-dominant gels (G' > G") within their operating frequency and strain ranges (Pi et al. 2023). The absolute magnitude of G′ under reservoir conditions serves as a measure of gel strength. (Schuman et al. 2022) developed a novel ultrahigh-temperature-resistant PPG (DMA-SSS PPG) that exhibited a G' of approximately 3200 Pa when fully swollen in formation water. The high G" value reflects a stiff gel network, which is attributed to the rigid divinylbenzene (DVB) crosslinks and strong polymer-polymer interactions. On the other hand, recrosslinkable or delayed-gelling particle systems are maintained at low modulus to ensure proper injectivity. (D. Y. Zhu et al. 2022) developed a recrosslinkable dispersed particle gel (RDPG) that remains a soft suspension with G' < 10 Pa under surface conditions.

One of the main benefits of the novel PPG formulations is their ability to tolerate high temperatures and salinity without severe degradation. Conventional in situ polyacrylamide-based gels hydrolyze and lose strength at 80-120 ˚C, especially in the presence of salts. Recent studies have improved the temperature resistance of PPG by modifying its chemical structure using heat-stable monomers and crosslinker. (Schuman et al. 2022), for example, reported that DMA-SSS PPG remained stable after long-term exposure at temperatures up to 130-150 ˚C. The salinity resistance was also significantly improved. (Almakimi et al. 2024; Azimifar, Tabatabaei-Nezhad, and Khodapanah 2025) reported that gels containing AMPS and NaSS monomers maintained their swelling capacity and structure in brines with TDS above 100 g/L. In some cases, reported by (Alotibi et al. 2024), higher salt concentrations increased gel stiffness, likely due to reduced swelling and tighter network packing.

Recent advances in gel-based conformance control have underscored the need for a more predictive, quantitative understanding of gel performance under reservoir conditions. While laboratory-scale studies provide insights into PPG behavior across specific temperature, salinity, and mechanical regimes, translating these findings into field-ready recommendations remains challenging due to the complex, nonlinear interactions between multiple formulation parameters and environmental conditions (Seidy Esfahlan, Khodapanah, and Tabatabaei-Nezhad 2021). Despite the fact that there a lot of mathematical modeling and simulations, they are focused mainly on PPG performance and propagation in the porous media and fractures (Leng, Wei, and Bai 2021; Seidy Esfahlan, Khodapanah, and Tabatabaei-Nezhad 2021). The traditional trial-and-error approach to gel design is time- and resource-intensive, especially for high-temperature and high-salinity reservoirs where gel degradation and injectivity failure are common. Additionally, there is a growing demand for predictive tools that can generalize across broad operating envelopes and rapidly screen candidate formulations without relying exclusively on expensive laboratory tests.

In this regard, machine learning methods offer a promising pathway to address this challenge by leveraging historical experimental datasets to model viscoelastic parameters such as G' and G". Despite increasing interest in ML-based material property prediction in other fields, the application of such tools to the systematic design and evaluation of PPGs remains limited. The recent examples of applying ML-based tools are still focused on field screening application of using PPG or predicting the wells injectivity (Aldhaheri et al. 2017; Aldhaheri, Bai, and Wei 2025).

The lack of integrated, physics-aware ML frameworks capable of handling multivariate datasets and capturing regime-specific presents a critical gap in the literature. By developing robust ML models for rheological characterization of PPGs, this work aims to bridge the gap between empirical experiments and practical design optimization for conformance control.

Methodology

Data description and analytics.

The dataset used in this study was acquired from experimental rheology studies provided by Kishor Pandit et al. 2024; J. Deng et al. 2024; Y. Zhu et al. 2024; J. N. Deng et al. 2025). The influencing parameters are frequency in Hz, strain in %, temperature in ˚C, salinity in %, particle size in micrometer and PPG type. The data for G' and G" were divided into two subsets and were separately modeled. The G' dataset contains 1446 datapoint for each parameter (total 10122 points). The G" dataset contains 297 datapoint for each parameter (total 2079 points). Additionally, G', G", frequency, and strain were log-transformed as these values are usually plotted in log-log scale. Both datasets were divided into 80:20 ratios for training and testing purposes, respectively. The statistical analysis of the training datasets is shown in Tables 1 and 2, and it contains the count, minimum, maximum, standard deviation, mean. There are 9 types of PPGs for the G' dataset and 8 types of PPGs for the G" dataset was used and coded as follows: 1 – WRPG, 2 – NRPG, 3 – MC-PPG, 4 – PPG without VSNPs, 5 – Pure PAM PPG/PAM-PPG, 6 – CR-PPG, 7 – CR-PPG without VSNPs, 8 – CR-PPG without crosslinking by Ca^{2+}, 9 – HSPPG. All coded types were assigned according to the original works.

Table 1—Statistical analysis of training dataset for G'.

	ε, %	Log(ε), %	f, Hz	Log (f), Hz	T, ˚C	S, %	Particle size, µm	G', Pa	Log (G'), Pa
count	1156	1156	1156	1156	1156	1156	1156	1156	1156
mean	97.08	0.36	4.78	0.13	66	2.81	967.13	32610.69	3.93
std	607.05	1.03	14.35	0.51	44.14	6.88	299.42	49064.97	0.76
min	0.0001298	-3.89	0.025	-1.60	20	0	200	160.81	2.21
max	5 663.16	3.75	100.53	2.0023	120	30	2000	433678.75	5.64

Table 2—Statistical analysis of the training dataset for G".

	ε, %	Log(ε), %	f, Hz	Log (f), Hz	T, ˚C	S, %	Particle size, µm	G", Pa	Log (G"), Pa
count	237	237	237	237	237	237	237	237	237
mean	23.10	0.10	1.20	-0.0032	23.33	0.77	871.31	3667.67	2.96
Std	84.79	1.40	1.30	0.29	2.36	1.31	219.06	7818.08	0.77
min	0.0001298	-3.89	0.02	-1.61	20	0	500	29.80	1.47
max	840.56	2.92	15.43	1.19	25	3	1000	59039.25	4.77

The correlation heatmap (Fig. 1) based on logarithmically transformed variables offers a clearer perspective on the relationships between experimental inputs and the logarithmic storage modulus, log(G'). The most prominent observation is the strong negative correlation between temperature and log(G'), with a coefficient of approximately −0.70. This indicates that as temperature increases, the gel's elasticity significantly decreases, likely due to thermal degradation or softening of the crosslinked polymer network.

This relationship highlights temperature as a critical factor in governing the rheological behavior of preformed particle gels.

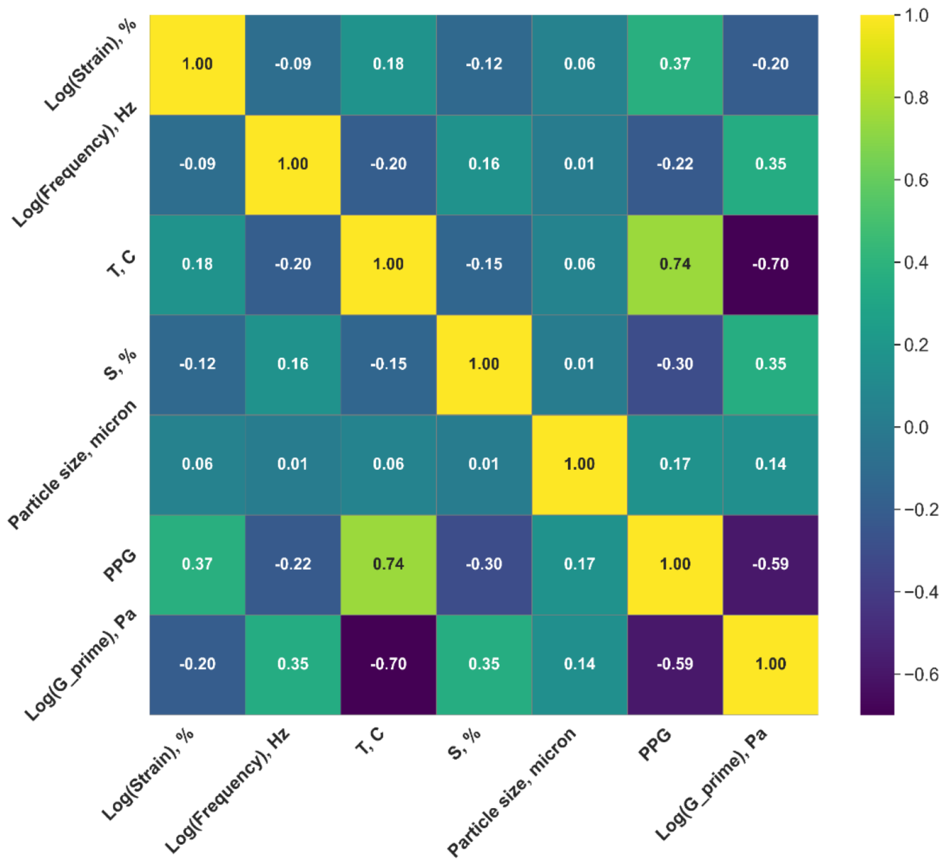

Figure 1—Correlation heat map for G'.

In addition to temperature, PPG type also shows a notable negative correlation with log(G') (around −0.59), implying that beyond a certain threshold, increasing gel content may actually reduce gel strength, possibly due to over-crosslinking or internal phase separation. On the other hand, both salinity and log(f) exhibit moderate positive correlations with log(G') (0.35 each), suggesting that higher ionic strength and higher testing frequencies contribute to enhanced gel stiffness. These effects may be attributed to increased ionic crosslinking and strain hardening behavior under oscillatory stress.

Other variables, such as log(ε) and particle size, show weak correlations with log(G') (−0.20 and 0.14, respectively), indicating limited linear influence on the logarithmic rheological response. Overall, the log transformation retains the major trends seen in the raw-scale analysis but sharpens the contrast, especially for the negative impact of temperature, reinforcing its dominant role in determining viscoelastic stability under elevated conditions.

The correlation analysis (Fig. 2) between log(G") and the other experimental variables provides insight into the key factors affecting the viscous behavior of preformed particle gels. The most significant positive correlation is observed with log(ε) (0.56), indicating that as strain increases, the viscous response of the gel becomes more pronounced. This suggests that under larger deformations, the gel dissipates more energy, reflecting greater internal friction or structural rearrangement.

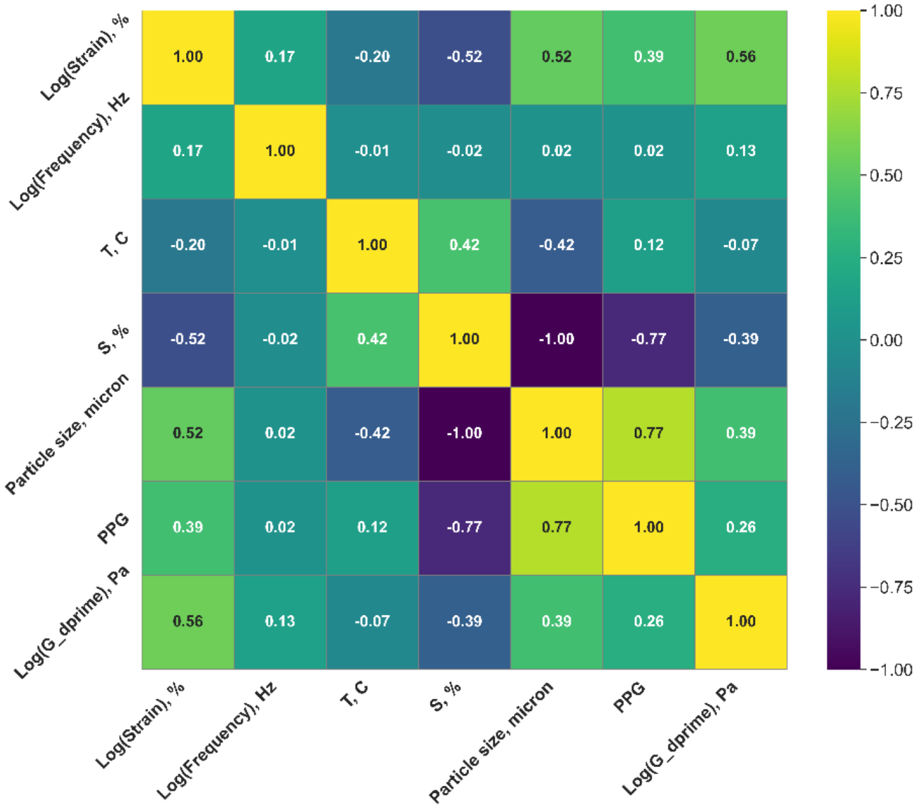

Figure 2—heat map for G".

A moderate positive correlation is also seen with particle size (0.39), implying that gels composed of larger particles tend to exhibit higher viscous resistance, possibly due to increased surface interactions or entanglement effects. PPG type shows a weaker but still positive correlation (0.26), indicating that different additives slightly enhance the loss modulus. In contrast, salinity shows a moderate negative correlation with log(G") (−0.39), suggesting that increased ionic strength suppresses the viscous component, potentially through ionic screening that limits polymer chain mobility.

Other variables such as log(f) and temperature exhibit weak correlations (0.13 and −0.07, respectively), indicating that their influence on the viscous modulus is minimal within the studied range. Overall, the most impactful variables on log(G") are strain, particle size, and salinity, with PPG concentration contributing to a lesser extent.

Modeling tools: Decision Tree, Gradient Boosting Regressor, Random Forest, and XGBoost.

Different ML tools were used in this work, namely Decision Tree, Gradient Boosting Regressor, Random Forest, and XGBoost. These models are well-suited for modeling G’ and G" because they handle nonlinear relationships and variable interactions. Such error metrics are the root mean squared error (RMSE), coefficient of determination (R^2), Akaike information criterion (AIC), and Bayesian information criterion (BIC).

For G’ (Table 3), the Decision Tree model exhibits the lowest RMSE (0.04 Pa) and the highest R^2 value (99.73%) for the training data, indicating an almost perfect fit. However, DT is known to overfit, which is reflected in its relatively weaker generalization on the testing data.

Table 3—Error metrics comparing the performance comparison of all applied ML tools for G'.

Data	Value	DT	GBR	RF	XGBoost
Training data	RMSE, Pa	0.04	0.06	0.06	0.06
	R^2, %	99.73%	99.45%	99.36%	99.46%
	AIC	–7474.74	–6649.38	–6471.70	–6651.35
	BIC	–7439.37	–6614.01	–6436.33	–6615.98
Testing data	RMSE, Pa	0.08	0.07	0.08	0.06
	R^2, %	99.09%	99.21%	98.93%	99.33%
	AIC	–1487.99	–1531.47	–1441.92	–1579.40
	BIC	–1462.30	–1505.78	–1416.23	–1553.71

In contrast, XGBoost, while having a slightly higher training RMSE (0.06 Pa), demonstrates the best performance on the testing data. It achieves the lowest RMSE (0.06 Pa), the highest R^2 (99.33%), and the most favorable information criteria values (AIC = −1579.40 and BIC = −1553.71). These results suggest that XGBoost provides a more generalizable model with a strong balance between accuracy and complexity. GBR also performs robustly, particularly in terms of R^2 (99.21%) and information criteria on the testing data, but XGBoost still surpasses it slightly. Random Forest, while comparable, ranks slightly lower across all metrics.

For G" (Table 4), In the training phase, the Decision Tree model demonstrates the best fit with the lowest RMSE (0.0016 Pa) and the highest R^2 value (99.99%). However, such a near-perfect fit typically indicates overfitting, as also seen in its relatively weaker performance on the testing data. XGBoost, despite having the highest RMSE in the training set (0.05 Pa) and slightly lower R^2 (99.64%), demonstrates more robust generalization. On the testing data, XGBoost achieves the lowest RMSE (0.09 Pa) and the highest R^2 (98.53%) among the models, suggesting better predictive performance on unseen data.

Table 4—Error metrics comparing the performance comparison of all applied ML tools for G".

Data	Value	DT	GBR	RF	XGBoost
Training data	RMSE, Pa	0.0016	0.02	0.02	0.05
	R^2, %	99.99%	99.92%	99.96%	99.64%
	AIC	–3023.89	–1793.93	–1970.05	–1439.96
	BIC	–2999.62	–1769.66	–1945.77	–1415.69
Testing data	RMSE, Pa	0.10	0.10	0.10	0.09
	R^2, %	98.28%	98.16%	98.14%	98.53%
	AIC	–262.85	–258.80	–258.25	–272.21
	BIC	–248.19	–244.14	–243.59	–257.54

In terms of model complexity and penalty-based metrics, XGBoost also outperforms the others in the testing dataset, recording the most favorable AIC (−272.21) and BIC (−257.54) values. These information criteria highlight the balance between model fit and complexity, reinforcing the superiority of XGBoost's generalization ability. Although the Decision Tree appears optimal on training data, its generalization is weaker due to likely overfitting, while GBR and RF perform similarly but fall slightly behind XGBoost in all testing metrics.

In summary, XGBoost consistently emerges as the most reliable and generalizable model for predicting both rheological moduli – elastic (G') and viscous (G"). For G', it demonstrates superior performance across testing metrics, particularly in terms of RMSE, R^2, AIC, and BIC, confirming its robustness on unseen data despite slightly higher training error compared to Decision Tree. Similarly, for G", XGBoost offers the best trade-off between accuracy and model complexity, outperforming other ensemble models such as GBR and RF, and significantly surpassing the overfitted Decision Tree model in generalization. The consistent dominance of XGBoost across both targets highlights its suitability for capturing the complex nonlinear relationships in the viscoelastic behavior of preformed particle gels.

Results and discussion.

Hyperparameters tuning.

The hyperparameters for each ML model were carefully tuned to optimize the prediction accuracy for both log-transformed G’ and G" (Table 5). For the Decision Tree, the maximum depth was optimized to 12 for log(G’) and log(G"), indicating the probable need for a deeper tree to capture more complex patterns. The minimum number of samples required to split an internal node was set to 3 and 2, and the number of samples at a leaf node was set to 1 for both, respectively. No restrictions were applied to the number of features considered at each split (None for both cases). In the GBR model, which builds models sequentially to reduce residuals, the number of estimators was set to 40 for both outputs. Both log(G’) and log(G") used a maximum tree depth of 7, while the minimum number of samples to split and to form a leaf node were 5 and 1 for log(G’) and log(G"), respectively. The model used "log2" for maximum features, ensuring that a subset of features was sampled at each split for regularization. For the RF, an ensemble of parallel trees of 200 estimators was used for both target parameters. The optimal maximum depth was 10, and both models used 2 for minimum samples split and 1 for minimum samples leaf. The maximum features were tuned to "log2" for G’ and "sqrt" for G", with bootstrap disabled in both cases, meaning the better performance with the full dataset rather than sampling with replacement. Last and most efficient model, XGBoost, was optimized with 100 estimators, a learning rate of 0.1, and a maximum depth of 7 for both cases. The minimum child weight was 1 for G’ and for G". The remaining parameters were fixed at a default values.

Table 5—Statistics of hyperparameters for all applied ML tools in this study.

ML tool	Hyperparameter	Range/Value For log(G’) and log(G")	Optimized value for log(G’)	Optimized value for log(G")
DT	Max depth	4-30	12	12
	Min samples split	2-20	3	2
	Min sample leaf	1-10	1	1
	Max features	sqrt, log2, auto, None	None	None
GBR	Number of estimators	20-40	40	40
	Max features	log2, sqrt	log2	log2
	Max depth	3-7	7	7
	Min samples were split	1-5	5	5
	Min sample leaf	1-3	1	1
RF	Number of estimators	100-300	200	200
	Max features	log2, sqrt	log2	sqrt
	Max depth	4-10	10	10
	Min samples split	2-10	2	2
	Min sample leaf	1-4	1	1
	Boostrap	True, False	False	False

ML tool	Hyperparameter	Range/Value For log(G') and log(G")	Optimized value for log(G')	Optimized value for log(G")
XGBoost	Number of estimators	25-100	100	100
	Max depth	5-9	7	7
	Learning rate	0.01-0.1	0.1	0.1
	Min child weight	1-5	1	3

Trend analysis.

Fig. 3 presents scatter plots comparing predicted versus actual values of the G′′ and G" using four machine learning models across both training and testing datasets. Subplots (a) and (b) correspond to G', while (c) and (d) represent G". The red dashed diagonal line in each plot represents the ideal case where predicted values match the actual ones perfectly, providing a visual benchmark for model accuracy and bias.

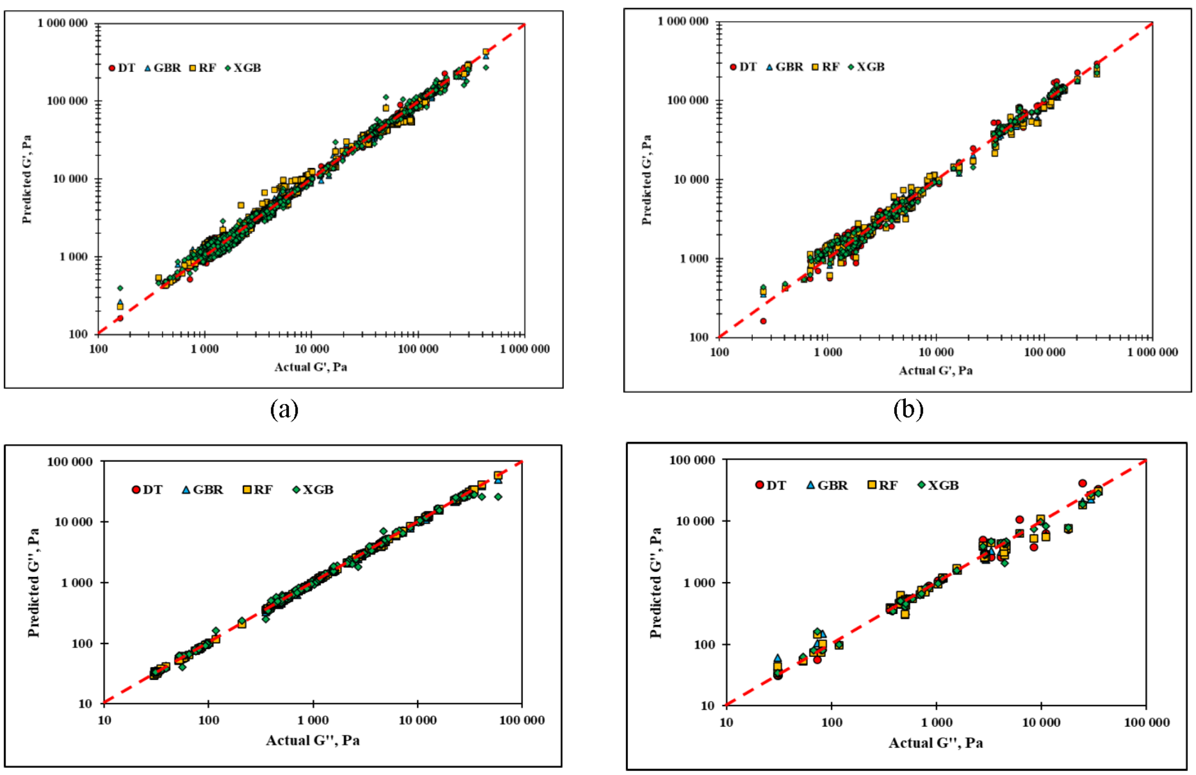

Figure 3—A comparison of scatter plots for all applied ML tools using (a) training dataset and (b) testing dataset for G' and (c) training dataset and (d) testing dataset for G".

In subplot (a), which illustrates predictions of G' on the training dataset, all models closely align with the diagonal, indicating strong fitting performance. The points are densely clustered around the line across several orders of magnitude, which is expected given the logarithmic scale. However, some deviations can be observed in DT and RF models at the higher modulus range, suggesting slight overfitting or bias. XGBoost and GBR maintain tight clustering and minimal dispersion, confirming their ability to capture the complex non-linear behavior of G' effectively.

Subplot (b) shows predictions on the testing dataset for G', where differences in model generalization become more apparent. While all models still follow the general trend, XGBoost predictions remain the most closely aligned with the diagonal, exhibiting minimal spread. In contrast, the DT model shows more scatter, particularly at both ends of the modulus spectrum, which supports the earlier observation of its overfitting.

GBR and RF perform reasonably well, though with slightly wider dispersion compared to XGBoost. This reinforces the earlier conclusion that XGBoost offers the best generalization on unseen data.

For G", subplot (c) (training data) reveals near-perfect predictions from all models, as the points lie almost exactly along the diagonal. The performance is particularly impressive given the wide dynamic range of G ″G″G″, from around 10 to 100,000 Pa. Nevertheless, the DT model shows a few points overshooting the diagonal, which could again be an indication of overfitting. XGBoost and GBR, by contrast, remain well-aligned across the entire modulus spectrum, suggesting more stable behavior during training.

Subplot (d) (testing data for G") provides the clearest differentiation in model generalization. Here, XGBoost predictions are most consistently close to the diagonal, demonstrating both accuracy and robustness. GBR and RF also follow the expected trend but exhibit slightly more scatter, especially at lower modulus values. The DT model again deviates more noticeably from the ideal line, confirming its tendency to overfit and underperform on independent data.

Overall, these scatter plots visually validate the quantitative metrics presented in earlier tables. XGBoost consistently delivers the most accurate and stable predictions for both G' and G" across training and testing datasets. Its performance is closely followed by GBR and RF, while DT, despite fitting well during training, demonstrates weaker generalization capability. These results confirm XGBoost as the most reliable model for capturing the viscoelastic behavior of preformed particle gels in this study.

Fig. 4 presents the frequency-dependent behavior of the elastic modulus G' for preformed particle gels (PPGs) at five temperatures (20 ˚C, 40 ˚C, 60 ˚C, 80 ˚C, and 100 ˚C). Colored dots represent experimental values compiled from literature source, while the dashed lines correspond to predictions made by the XGBoost model. This comparison enables a comprehensive assessment of the model's capacity to replicate actual rheological behavior in both the thermal and frequency domains.

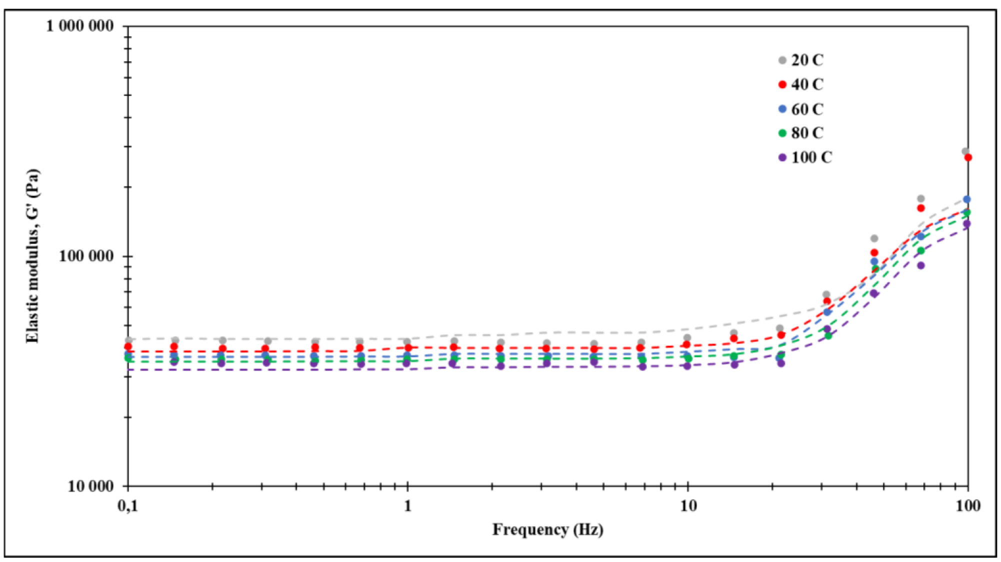

Figure 4—Effect of temperature on G' (dots are experimental data points and colored dashed lines show the predicted trend based on XGBoost)

Overall, the XGBoost predictions show excellent agreement with the experimental data. Across all temperatures, the model captures the general viscoelastic trend where G' remains nearly constant at low frequencies and then increases significantly beyond a critical frequency threshold. In the low-frequency range (approximately 0.1 to 10 Hz), the predicted curves align closely with the experimental points, reflecting the gel's plateau-like elastic behavior. At higher frequencies, particularly above 20 Hz, the model continues to trace the data well, preserving the steep rise in modulus that corresponds to increased rigidity under rapid deformation.

However, two notable deviations are observed. First, at 20 °C, the grey dashed line begins to diverge from the experimental data beyond 1 Hz. While the model successfully captures the low-frequency plateau, it slightly underestimates the sharp increase in modulus observed experimentally at higher frequencies. This discrepancy may be attributed to limited training data density for this specific temperature-frequency regime, or possibly to subtle physical transitions (i.e., secondary network stiffening or structural rearrangements) at 20 ˚C that are not fully represented in the features learned by the model.

Second, for the other temperatures (40 ˚C to 100 ˚C), the predicted curves begin to slightly deviate from the real data after approximately 20 Hz. Although the predicted trends maintain the correct shape and overall magnitude, they tend to slightly underestimate the actual modulus values at the highest frequencies. This behavior likely arises from two factors: (1) the scarcity of high-frequency experimental data at elevated temperatures in the training set, and (2) the challenge of capturing sharp nonlinear transitions in highly dynamic regimes using a tree-based model, which excels in capturing smooth nonlinear patterns but may over-smooth sudden transitions unless well-represented in the data.

Fig. 5 illustrates the relationship between the elastic modulus G' and applied strain for preformed particle gels (PPGs) with varying particle sizes: 200 µm, 500 µm, 1000 µm, and 2000 µm. The x-axis (strain, %) and y-axis (modulus, Pa) are both presented on logarithmic scales, covering a wide deformation range from 0.1% to 100%. The colored markers represent actual experimental values extracted from literature sources, while the dashed lines indicate the predicted behavior generated by an XGBoost regression model trained on rheological features.

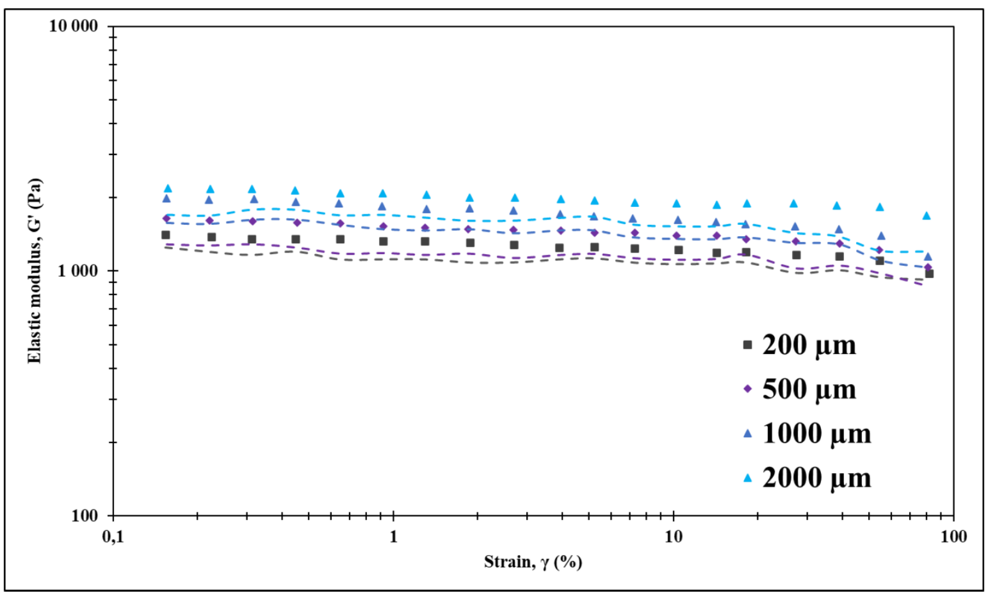

Figure 5—Effect of particle size on G' (dots are experimental data points and colored dashed lines show the predicted trend based on XGBoost)

Across all particle sizes, the elastic modulus G' remains relatively stable with increasing strain, suggesting that the PPG systems maintain their structural integrity and resist deformation within the tested strain window. This is characteristic of materials exhibiting a linear viscoelastic plateau, where the internal gel network structure remains largely intact before yielding or structural breakdown. Among the particle sizes tested, PPGs with larger particles (1000 µm and 2000 µm) consistently demonstrate higher modulus values compared to smaller particles (200 µm and 500 µm), implying that larger particles form a more rigid, interconnected structure with enhanced elastic resistance under strain.

The XGBoost-predicted trends (dashed lines) generally capture the correct magnitude and qualitative trend of the elastic modulus across all particle sizes. The model correctly identifies the relative ranking between particle sizes and preserves the expected plateau shape. However, it is evident that the predictions do not align perfectly with the experimental data points. Specifically, the dashed lines tend to slightly underestimate G' at all strain intervals, and some local fluctuations in the experimental data are not fully reflected in the predicted curves.

These deviations may be attributed to several factors. First, the experimental data used in training may have limited representation of certain particle sizes or strain regimes, which can reduce the model's ability to generalize fine-scale variations. Second, experimental inconsistencies across literature sources, such as differences in additives concentration, crosslinkers, can introduce variability not captured by the model's features. Finally, XGBoost, being a tree-based model, tends to approximate nonlinear relationships through piecewise partitions, which may smooth out local oscillations or noise inherent in empirical datasets.

In conclusion, this plot confirms that XGBoost is capable of capturing the dominant physical trends related to the effect of PPG particle size on elastic modulus across strain levels. Nevertheless, mismatches between predicted and actual values highlight the need for either broader training data coverage or incorporation of more refined features (i.e., network density, swelling ratio) to enhance predictivity in future model iterations.

Fig. 6 illustrates the impact of salinity on the elastic modulus G' of preformed particle gels (PPGs) as a function of oscillatory frequency, ranging from 0.1 to 100 Hz. Experimental data points (dots), gathered from publications, are compared with predictions (dashed lines) generated using an XGBoost regression model trained on a compiled dataset. Four salinity conditions are analyzed: DI water (green), 10% NaCl (blue), 30% NaCl (red), and a binary divalent system of 10% $CaCl_2$ + 10% $MgCl_2$ (pink).

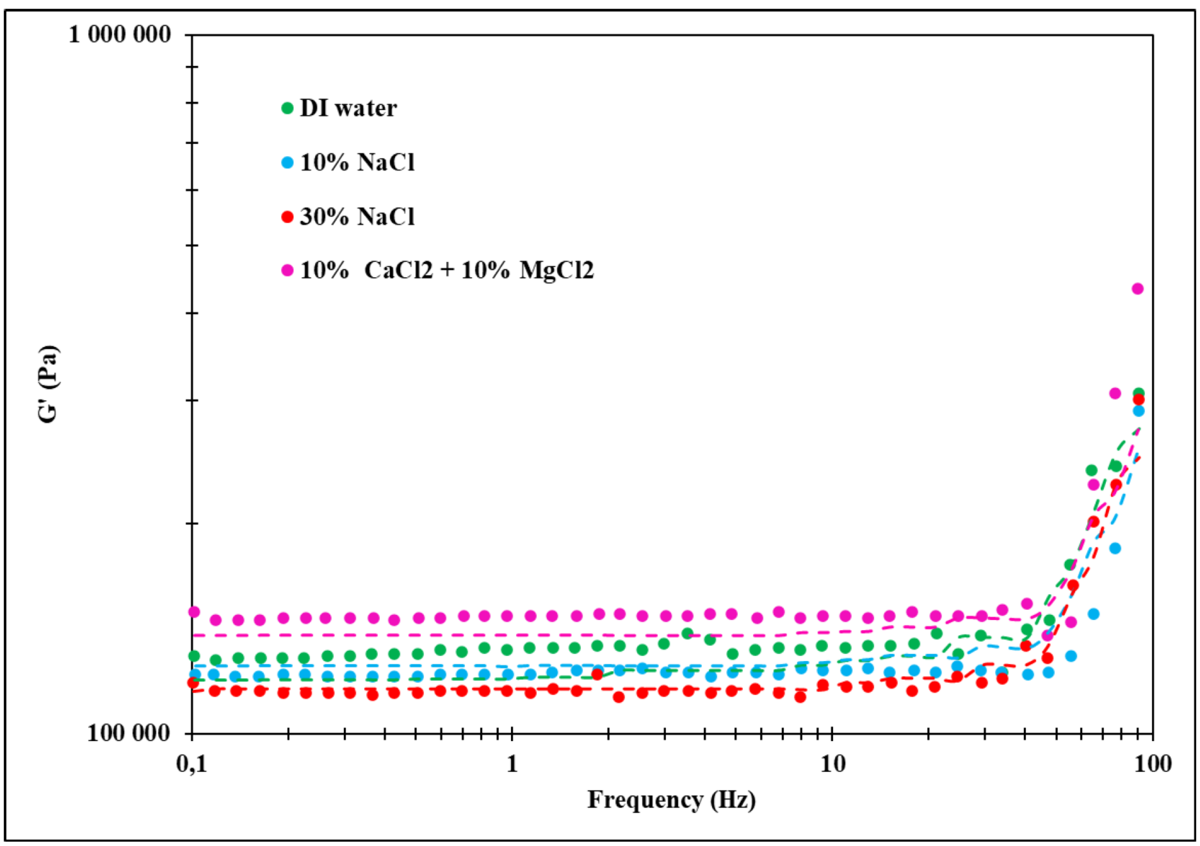

Figure 6—Effect of salinity on G' (dots are experimental data points and colored dashed lines show the predicted trend based on XGBoost)

Overall, the elastic modulus G' exhibits typical viscoelastic behavior under all salinity conditions: it remains nearly constant in the low-frequency regime and increases sharply beyond approximately 20

Hz, corresponding to the transition from linear to nonlinear viscoelastic response. Among the tested environments, the highest modulus is observed for the divalent salt system (pink), followed by DI water (green), 10% NaCl (blue), and 30% NaCl (red), demonstrating that both ionic strength and ion valency significantly influence gel stiffness.

The XGBoost predictions generally align well with experimental values in terms of magnitude and trend, especially in the intermediate frequency range (~1 to 10 Hz). However, several important deviations are evident upon closer inspection. Notably, the pink dashed line (10% $CaCl_2$ + 10% $MgCl_2$) fails to capture the real trend across the entire frequency range. While it reproduces the general order of magnitude, it underestimates the modulus at both low and high frequencies and misses the actual curvature of the dataset, indicating the model's limited ability to extrapolate in the presence of strong multivalent ion effects.

Similarly, the green dashed line (DI water) remains slightly below the experimental data from 0.1 to 20 Hz and does not reproduce the fine-scale fluctuations observed between approximately 3 and 6 Hz. This suggests that while the model approximates the overall plateau, it smooths over localized variations that may be caused by subtle structural rearrangements in the gel network.

The blue dashed line (10% NaCl) is consistently slightly above the experimental data throughout the entire frequency range and also fails to capture the actual curvature of the modulus. This systematic deviation indicates that the model may have overgeneralized based on similar NaCl cases in the training set, leading to reduced accuracy in representing this specific formulation.

The red dashed line (30% NaCl), while relatively close in trend, does not reproduce the localized fluctuation observed in the experimental data between approximately 1.7 and 2.2 Hz. This local deviation may be linked to concentration-dependent microstructural effects or gel softening mechanisms that are not sufficiently represented in the training features.

In summary, while XGBoost performs well in approximating the general magnitude and ranking of G' across different saline conditions, it struggles to capture local fluctuations and detailed curvature in several cases. The pink and blue curves deviate from the real trends across the entire frequency spectrum, while the green and red curves exhibit localized inaccuracies in specific frequency intervals. These mismatches may arise from the smoothing nature of tree-based models, lack of fine-grained training data in certain regimes, or underlying physicochemical mechanisms, such as ion-specific interactions, partial crosslinking, or gel restructuring, that are not explicitly included as input features.

Fig. 7–8 illustrates the variation of G' and G" with shear strain for different types of preformed particle gels (PPGs), showcasing the effect of gel composition and nanomaterial inclusion on viscoelastic behavior.

Fig. 7, the focus is on two gel types: WRPG and NRPG, each with its corresponding G' and G" curves. For both G' and G", the WRPG maintains a higher plateau at low strains and undergoes a pronounced drop beyond approximately 0.1% strain, indicative of yielding behavior. The NRPG shows a similar trend, albeit starting from lower initial moduli. The XGBoost model effectively captures the overall decline in G' for both systems as strain increases, as well as the relative stiffness between WRPG and NRPG. However, certain deviations are apparent: for WRPG (black curves), the predicted G' curve starts diverging from the experimental data beyond ~0.05% strain, slightly underestimating the steepness of the modulus drop. Similarly, for NRPG (red curves), the predicted G" values do not fully reproduce the local curvature in the transition region between 0.1% and 10%, indicating a smoothing effect typical of tree-based models that can obscure sharp nonlinear transitions.

Fig. 8 examines G' and G" behavior for four systems: MC-PPGs, PPGs without VSNPs, and pure PAM PPGs, with and without nanoparticles. All systems exhibit the expected viscoelastic pattern, with a plateau at low strain followed by a decline at higher strain values, characteristic of structural breakdown or yielding. MC-PPGs (green and cyan) show the highest G' values, indicating enhanced rigidity due to nanomaterial reinforcement. The XGBoost model captures these general trends and ranks the systems correctly in terms of stiffness. However, for the pure PAM PPGs (red), the predicted G" curve notably deviates from the experimental data in the intermediate strain range (~2–10%), failing to capture the increase and peak-like structure before the drop. Likewise, for PPGs without VSNPs (blue), the predicted G' line begins to diverge after about 20% strain, underestimating the gradual decline observed in the real data.

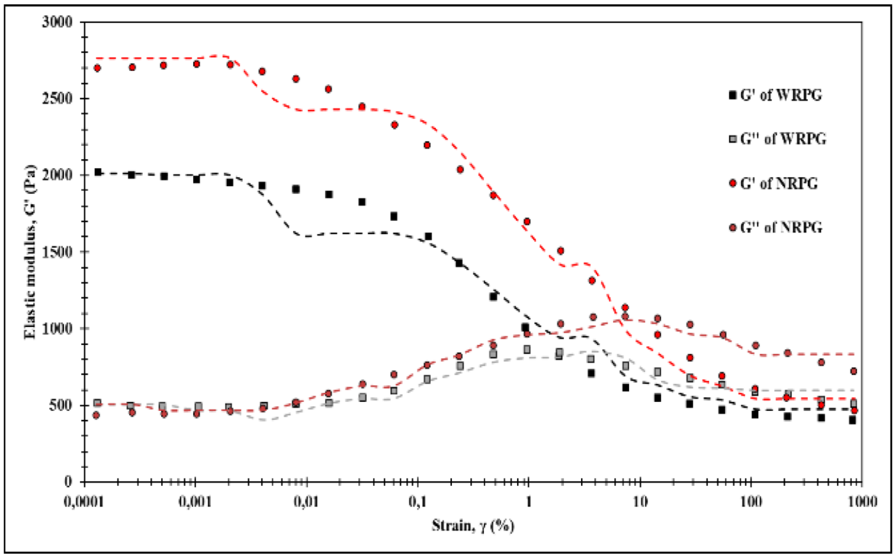

Figure 7—Effect of PPG type on G' and G" (dots are experimental data points and colored dashed lines show the predicted trend based on XGBoost), (Kishor Pandit et al. 2024)

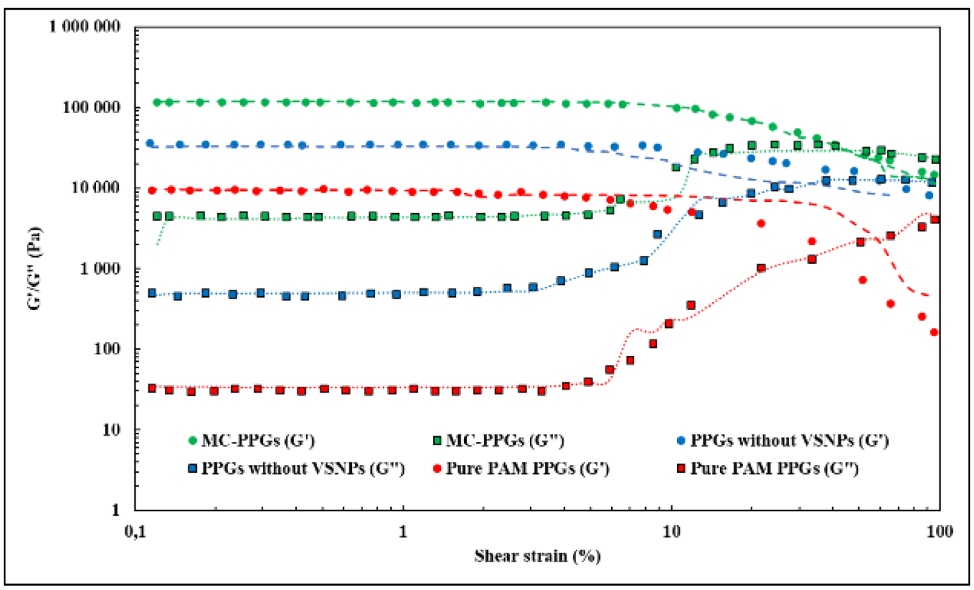

Figure 8—Effect of PPG type on G' and G" (dots are experimental data points and colored dashed lines show the predicted trend based on XGBoost), (J. Deng et al. 2024)

These mismatches can be attributed to several factors. First, XGBoost relies on discrete partitioning of feature space and may smooth over sharp transitions or local fluctuations in data-rich nonlinear regions. Second, the experimental data come from different sources with varied formulation details, such as nanoparticle type, concentration, and crosslinking mechanism, which may not be fully reflected in the input features. Third, noise in the original data, especially around critical transition zones, can cause minor inconsistencies that the model tends to average out.

In conclusion, XGBoost performs well in predicting the broad viscoelastic behavior of different PPG systems, accurately identifying major trends and relative magnitudes of G' and G". However, certain localized nonlinear features, such as strain-induced peaks, inflection zones, or abrupt transitions, are not perfectly captured in specific regions, particularly between 0.1% to 10% strain, due to the model's inherent smoothing and data limitations. These results reinforce the promise of ML in rheological prediction, while also highlighting the importance of richer, more feature-dense datasets and possibly hybrid modeling strategies to better resolve sharp viscoelastic transitions across PPG chemistries.

Conclusions

In this study, various machine learning models, including Decision Tree, Gradient Boosting Regressor, Random Forest, and XGBoost, were employed to predict the viscoelastic properties of preformed particle gels (PPGs), namely the elastic modulus (G') and the loss modulus (G"). Among them, XGBoost consistently demonstrated the best generalization ability, with the lowest RMSE (0.06 and 0.09) and the most favorable AIC (-1579.40 and -272.21) and BIC (-1553.71 and -257.54) values for both G' and G" on test datasets, respectively. However, its performance advantage over other ensemble models, such as the RF and GBR models, was relatively small, suggesting that all three models are suitable for high-quality regression of rheological parameters.

Despite its strong performance in global trend prediction, the XGBoost model showed limitations in capturing localized nonlinear behaviors and structural transitions, such as softening at high strain, sharp thermal responses, and yielding behavior. This was particularly evident in comparisons between the model predictions and experimental data, where XGBoost tended to oversmooth the curves and misrepresent the critical points of inflection under varying thermal, mechanical, and chemical conditions. These findings highlight the strengths of tree-based models in learning complex, multivariate relationships and emphasize the need for hybrid approaches or physics-informed models when modeling materials with regime-specific transitions.

A key novelty of this study lies in its integration of experimental rheological data from diverse literature sources into a unified supervised learning framework, enabling quantitative comparison between machine learning predictions and known viscoelastic trends across varying PPG chemistries, temperatures, salinities, and strain levels. By directly benchmarking model performance against experimental transition behaviors, this study bridges the gap between data-driven algorithms and domain-specific physical understanding.

In future studies, it will be recommended to explore hybrid modeling strategies that combine machine learning with mechanistic or physics-informed features, particularly for better capturing nonlinearity near yield points. Additionally, incorporating uncertainty quantification, model interpretability techniques, and broader datasets, especially those covering high-temperature and high-salinity conditions, would further enhance the robustness and applicability of ML-based rheological modeling for advanced gel systems for conformance control operations.

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